

Progress Invoices, Change Orders, and Contract Invoices

A consolidated development requirements review of the current Cinderblock frontend prototypes, including contractor usability, UI quality, cross-flow risks, and decisions needed before production implementation.

Review basis

Current prototype requirements, local state models, implemented React flows, shared Cinderblock UI documentation, supplied internal screenshots, and customer-facing PDF/weblink references.

Review lens

The prototypes were evaluated as product-design artifacts. Grades reflect how confidently a contractor could understand and use the proposed workflows, and how ready the interaction model is for development review. They are not grades for production accounting correctness.

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1. Executive assessment

The three prototypes form a strong product direction: Progress Invoices cover estimate-only tracked billing, Change Orders preserve contract history while modifying scope, and Contract Invoices combine accepted Estimate and Change Order scope into one billable contract view.

The main risk is no longer screen design. It is the absence of one authoritative cross-flow contract and billing model. Several local prototype rules are individually plausible but conflict when the workflows meet - especially Draft consumption, invoice eligibility after Change Order acceptance, and the handoff from Estimate-only Progress Invoices to contract-wide invoicing.

Contractor workflow

B+

8.2 / 10

Clear and usable direction; missing a few field-critical rules.

Overall UI

A-

8.7 / 10

Feels like Cinderblock; dense contract tables need refinement.

Logic readiness

B-

7.4 / 10

Strong per-flow rules; cross-flow invariants remain unsettled.

What is working

- The Estimate remains the source contract document and the visual entry point for billing and modifications.
- Progress billing preserves all contract lines, distinguishes prior billing from the current draw, and supports percentage, amount, item selection, and remaining balance.
- Change Orders preserve source history, represent removals as negative quantity adjustments, support optional explicit replacements, and expose contract lineage clearly.
- Contract Invoices correctly group the accepted contract by source document and make contract-level progress authoritative in the internal editor.
- Customer previews explain both line-level completion and total contract progress while retaining historical context.

Top decisions before development

Priorit y	Decision	Why it blocks implementation
P0	Define the authoritative billing ledger and Draft semantics	Current prototypes alternately exclude Draft from Previously billed while including saved Draft allocations in contract consumption/progress.
P0	Define when Progress Invoice becomes Contract Invoice	After a CO is accepted, the product needs one unambiguous invoice path, one draft rule, and one remaining-balance calculation.
P0	Define contract mutation while an invoice Draft exists	An accepted CO can add, remove, or replace lines that a Draft already references.
P0	Define accepted-document edit and audit rules	Both accepted Estimates and accepted COs expose Save changes patterns, but production versioning consequences are unspecified.
P1	Define approval, cancellation, credit, and over-invoice resolution	The prototypes display these states but intentionally stop before operational and accounting resolution.

2. Required unified business model

Development should implement the three experiences on one contract ledger. UI modules may remain separate, but they must read and write the same domain state.

Core entities

Entity	Required responsibility
Contract	Owns the original accepted Estimate, accepted Contract value, contract status, parties, job, and currency.
Contract document	Estimate or Change Order with immutable identity, revision history, status, source document, and effective date.
Contract line	Stable line identity, source document, signed quantity/value, active/superseded state, and optional replacement relationship.
Invoice	Immutable invoice type, status, dates, customer terms, document links, totals, and payment state.
Invoice allocation	Links an invoice line amount to one billable contract line. This is the source of Previously billed and remaining scope.
Discount pool	Tracks accepted Estimate discount, applied allocations, remaining pool, and relationship to later contract changes.
Audit event	Records create, edit, send, accept, decline, cancel, void, delete, and status transition actor/time.

Authoritative invariants

- Contract value equals the accepted Estimate total plus the net delta of Accepted Change Orders only.
- Draft, Pending, Declined, Canceled, and Void Change Orders do not change accepted Contract value.
- Every accepted source document and contract line has a stable ID; names and display order are never used as identity.
- A removed source line stays in history, becomes non-billable, and is represented by a negative adjustment on the Change Order.
- A replacement is an explicit optional relationship. Independent additions have no replacement link.
- Previously billed is derived from qualifying historical invoice allocations and always excludes the invoice currently being viewed.
- Invoice type is fixed by the selected workflow. Billing 100 percent in a Progress/Contract Invoice does not convert it into a Standard invoice.
- No billable line may receive more new allocation than its remaining eligible amount unless an explicit overbilling policy permits it.
- Customer totals, internal totals, Estimate summaries, and Contract details must be projections of the same authoritative ledger.

Recommended status vocabulary

Object	Statuses	Counted in accepted contract / billing
Estimate	Draft, Sent, Accepted, Partially converted, Converted, Declined, Void	Accepted establishes original contract value.
Change Order	Unsaved, Draft, Pending, Accepted, Declined, Canceled, Void	Only Accepted affects Contract value.

Object	Statuses	Counted in accepted contract / billing
Invoice	Unsaved, Draft, Sent, Partially paid, Paid, Void	Product must explicitly decide whether Draft reserves availability, affects displayed progress, or both.

3. Progress Invoice requirements

Purpose: tracked billing against an accepted Estimate before accepted Change Orders expand or modify the contract.

Entry and creation

ID	Requirement
PI-01	An accepted, uninvoiced Estimate opens What type of invoice? with Standard invoice and Progress invoice.
PI-02	Standard invoice copies all Estimate lines into a flexible invoice and retains only a document-level Estimate link.
PI-03	The first Progress Invoice offers Percent or amount and Select items. It does not offer the rest of the estimate.
PI-04	A subsequent Progress Invoice also offers The rest of this estimate and displays the remaining amount.
PI-05	Percent mode applies the entered percentage across eligible Estimate scope; amount mode distributes a dollar amount across eligible scope.
PI-06	Selecting an item invoices its full remaining eligible amount. Fully invoiced items remain visible but disabled.
PI-07	Back preserves selections; Cancel or closing unsaved work creates no allocation and changes no Estimate state.

Discount pool

- If no discount remains, Continue opens the editor directly.
- If discount remains, the same creation dialog adds an Estimate discount step before the editor.
- The suggested proportional amount is editable, may be zero, and cannot exceed the remaining pool.
- Applied Draft and posted discounts need an explicit ledger rule; production must prevent the pool from being consumed twice.

Editor and lifecycle

ID	Requirement
PI-08	Every Estimate line appears in the editor, including zero-current lines. Columns are Item, Qty, Contract Amount, Previously Billed, This Invoice, and percent Complete.
PI-09	Qty, Contract Amount, and Previously Billed are read-only context. Percent Complete is derived from prior plus current allocation.
PI-10	Draft invoices are editable; non-Draft carried contract lines are locked.
PI-11	Only one Draft Progress Invoice may reserve scope for an Estimate at a time.
PI-12	Saving must preserve the invoice ID, amounts, discount, dates, notes, and terms. The product team must choose whether save remains on the invoice or returns to the Estimate; current prototype documentation and newer Contract Invoice behavior differ.
PI-13	Estimate status is Accepted before billing, Partially converted during progress billing, and Converted when all eligible Estimate value is consumed.
PI-14	Linked invoices appear in Estimate Details and reopen the exact saved invoice state.

ID	Requirement
PI-15	PDF-style and web previews use live invoice data, show all contract lines by default, and visually subdue zero-current lines.

Contractor assessment

Grade: **B+ (8.4/10)**. The choices map well to how contractors think about a draw: bill a percent, a dollar amount, selected scope, or the remainder. The all-lines editor is trustworthy because it keeps the whole estimate visible. The missing production decisions are material: retainage, deposits, stored materials, approved-but-unbilled changes, and an explicit Draft reservation explanation.

4. Change Order requirements

Purpose: preserve the original accepted contract while documenting additive, deductive, and replacement adjustments in a sequential contract lineage.

Creation and eligibility

ID	Requirement
CO-01	Create change order is available from accepted contract context when no Draft or Pending CO exists.
CO-02	Only one Draft or Pending CO may exist per contract. The disabled action explains that a CO is already in progress.
CO-03	Creation from an Accepted CO keeps that CO visible beneath the selector and uses its eligible positive lines as source scope.
CO-04	Source selection is optional. Addition-only, removal-only, replacement, and mixed COs are valid.
CO-05	A source line already reserved or adjusted by a Draft, Pending, or Accepted CO remains visible in selection context but cannot be selected again.
CO-06	A new unsaved CO opens with one blank editable line after any selected removal lines and has no status, Contract summary, or Contract details yet.

Adjustment model

Adjustment	Representation	Billing consequence
Removal	Original positive Cost/Markup/Price; negative Qty; negative Amount; locked and struck through in the CO.	The source line becomes non-billable after acceptance. The negative adjustment is never directly billable.
Replacement	One optional editable positive line nested under a removed line with an explicit relationship ID.	Future carryover of historical billing is unresolved and must not be inferred from the visual pairing.
Independent addition	Ordinary positive line added with Add another and no replacement relationship.	Becomes new billable contract scope only after the CO is Accepted.

Lifecycle

ID	Requirement
CO-07	Save and Send remain disabled until at least one meaningful non-zero adjustment exists. Net-zero is valid when real positive and negative adjustments offset.
CO-08	Save creates Draft. Send creates Pending. Acceptance moves the net delta into accepted Contract value.

ID	Requirement
CO-09	Draft may be deleted after confirmation; deletion removes history references, releases reserved source lines, and returns to the originating contract document.
CO-10	Pending may be canceled after confirmation; Canceled remains history, affects no accepted Contract value, and releases reserved source lines.
CO-11	Accepted, Declined, and Canceled COs remain in Contract details and cannot be deleted.
CO-12	Accepted COs expose the same contract action row as an accepted Estimate. The top document header does not show Create change order after save.
CO-13	Contract details lists the Estimate, sequential CO deltas and unsettled statuses, and Current contract. Accepted CO status is visually quiet.
CO-14	Contract summary uses accepted Contract value. A Draft/Pending delta appears as secondary proposed context without changing progress.
CO-15	An Accepted deductive CO may produce Over invoiced. Existing invoices remain unchanged; progress may exceed 100 percent and the bar distinguishes contract coverage from overage.

Contractor assessment

Grade: **B (8.0/10)**. The negative-quantity adjustment and visible lineage are strong. Optional replacement pairing is useful for later billing reconciliation. The workflow still needs the information contractors rely on when a customer disputes scope: approval date, signer, version issued, reason for change, schedule impact, and a clear revised-contract preview before sending.

5. Contract Invoice requirements

Purpose: tracked billing across the full accepted contract after one or more accepted Change Orders exist. It must include Estimate history, accepted CO adjustments, and current billable scope without rewriting historical allocations.

Creation and scope

ID	Requirement
CI-01	The accepted Contract value is the original Estimate plus Accepted CO deltas. Non-Accepted COs are excluded.
CI-02	Creation offers The rest of this contract, Percent or amount, and Select items when previous contract billing exists.
CI-03	Select contract items shows eligible current scope. Removed source lines and negative CO adjustments are omitted; fully invoiced current scope may remain visible but disabled.
CI-04	Percentage and amount allocation may target only eligible current lines and may not exceed contract remaining or line remaining.
CI-05	The editor groups all historical and current contract lines by source document, with repeated column headers beneath each Estimate or CO header.

Internal editor

ID	Requirement
CI-06	Columns are Item name, Qty, Contract, Previously billed, and This invoice. Internal line-level percent Complete is intentionally omitted.
CI-07	Contract progress is projected live from accepted Contract value, Previously invoiced, This invoice, and Remaining after invoice.

ID	Requirement
CI-08	Superseded Estimate lines retain historical billing and are non-billable. Negative CO adjustments show em dashes for prior and current billing.
CI-09	Replacement and new CO lines receive billing only through their own allocations; historical amounts are not silently transferred.
CI-10	Saving transitions the same editor from new to saved Draft, preserving values and keeping the user on the invoice document.
CI-11	Projected overbilling is displayed as Over invoiced with mathematical progress above 100 percent and a green/red normalized bar.

Customer presentation

- Each source repeats Source document header, compact column headers, then source line items.
- Customer columns remain Item, Qty, Contract amount, Previously billed, This invoice, and percent Complete.
- A removed original line stays readable with Removed by CO reference and its real historical billing. The matching negative adjustment is struck through and uses em dashes for billing.
- Invoice totals appear before the secondary Contract progress block. Contract progress shows Contract amount, Billed so far, and percent Complete.
- The web view defaults Show all contract items on; the PDF uses grayscale to subdue zero-current rows.

Contractor assessment

Grade: **B+ (8.3/10)**. Source grouping and overall progress solve a real problem after contract changes. The current label set is still too close to Progress Invoice, and the customer table is dense. Contractors need a clear rule for when this flow replaces Estimate-only Progress Invoices and how Draft draws behave when another CO is accepted.

6. Cross-flow requirements

These requirements are the minimum needed to keep the three workflows internally consistent.

ID	Requirement	Acceptance signal
XF-01	All flows use one Contract, ContractLine, Invoice, and InvoiceAllocation ledger.	The same Contract value, billed total, and remaining amount appear from Estimate, CO, and Invoice views.
XF-02	The system chooses one tracked billing path for a contract at a time based on contract composition.	After the first Accepted CO, the entry action and editor type are unambiguous.
XF-03	Invoice type is immutable and stored explicitly.	A 100 percent Progress or Contract Invoice remains that type when reopened.
XF-04	The one-Draft rule has a defined scope: Estimate, contract, or invoice type.	The UI never allows two drafts to reserve the same billable scope accidentally.
XF-05	Accepted CO processing recomputes eligibility from stable line IDs and never mutates historical allocations.	Old invoices reopen unchanged; new invoices reflect new current scope.
XF-06	CO acceptance checks for open invoice Drafts and follows a defined rebase, block, or review policy.	No Draft silently gains, loses, or reallocates lines after acceptance.
XF-07	Discount, tax, and Cost Plus rules identify whether they belong to the original Estimate, a CO, an invoice, or the contract.	Totals reconcile across internal editor and both customer previews.
XF-08	Linked to is reserved for invoice-document relationships; Contract details owns contract lineage.	Estimate and CO views never duplicate lineage in Linked to.

ID	Requirement	Acceptance signal
XF-09	Delete, cancel, decline, void, and acceptance transitions update reservations atomically.	Availability, lineage, totals, and actions update together.
XF-10	Customer-facing documents are immutable snapshots of the issued revision.	Reopening a historical link shows the issued content even after later contract changes.

Recommended tracked-invoice gateway

For production review, use a single Create invoice action from contract context. If the accepted contract contains only the original Estimate, offer Standard invoice or Progress invoice. Once any Change Order is Accepted, route tracked billing to the contract-wide Contract Invoice experience. Existing Estimate-only Progress Invoices remain historical allocations in the same contract ledger.

This recommendation removes a major user-facing ambiguity without merging the three prototype modules. Product should approve or reject it explicitly.

Calculation definitions

Metric	Definition
Accepted Contract value	Accepted Estimate total + sum of Accepted CO deltas.
Previously billed	Qualifying posted allocations before the current invoice; exact qualifying statuses require approval.
Reserved	Optional separate amount held by saved Draft allocations; must not be mislabeled as billed.
This invoice	Current unsaved or saved invoice allocations, excluding non-billable adjustments.
Remaining after invoice	Accepted Contract value - Previously billed - This invoice, floored at zero for display.
Over invoiced	$\max(0, \text{qualifying billed amount} - \text{Accepted Contract value})$.
Contract progress	Qualifying billed amount / Accepted Contract value; may exceed 100 percent in the over-invoiced state.

7. Blindspots and unresolved product decisions

P0 - decide before implementation

Blindspot	Current tension	Required decision
Draft semantics	Progress Invoice docs say prior billing excludes Draft, yet saved Draft allocation contributes to Estimate consumption/progress. Contract Invoice totals currently aggregate saved Drafts.	Define posted, reserved, billed, and paid as separate values and decide which each screen displays.
Tracked invoice transition	Progress Invoice and Contract Invoice are separate sandboxes with overlapping creation and editor language.	Define the exact trigger and migration when the first CO is accepted.
Open Draft plus CO acceptance	A CO may replace a line already present in an editable invoice Draft.	Block acceptance, rebase the Draft with user review, or preserve the Draft against the prior contract revision.
Accepted document edits	Accepted Estimate/CO patterns expose Save changes but issued contract documents should be auditable.	Choose immutable revisions, amendment workflow, or narrowly permitted metadata edits.

Blindspot	Current tension	Required decision
Replacement billing inheritance	CO stores replacement linkage, while Contract Invoice intentionally leaves historic billing on the old line.	Decide if and how billing credit follows replacement scope; define customer explanation and ledger entries.
Approval evidence	Pending and Accepted states exist without finalized signature/acceptance rules.	Define signer identity, timestamp, version hash, decline reason, resend, expiration, and manual acceptance permissions.

P1 - decide before pilot

- Retainage: whether it is contract-wide, line-level, released separately, and shown in customer totals.
- Deposits and prepayments: whether they count as billed, paid, or available credit against later draws.
- Stored materials and percent-complete evidence: whether contractors need quantity/work-complete notes or attachments by line.
- Partial removal: the current CO selection removes the full source quantity. Define quantity reduction and unit-rate changes.
- Tax, discount, and Cost Plus after COs: define whether accepted deltas inherit original settings or carry document-specific rules.
- Standard invoice coexistence: define whether Copy to invoice permanently exits tracked billing or may coexist with later COs.
- Over-invoiced resolution: the prototype is informational only; define credit memo, next-invoice offset, refund, or manual resolution policy.
- Void behavior: define how voiding an invoice or CO reverses reservations, billing totals, linked references, and customer history.

P2 - improve after core rules

- Field/mobile usability for long tables, item selection, approvals, and photo attachments.
- Role permissions, approval thresholds, and internal review for high-value or deductive COs.
- Search, export, job-level reporting, aging, and reconciliation across many invoices and COs.

8. UI and interaction review

Overall UI grade: **A- (8.7/10)**. The prototypes extend the existing Cinderblock visual language rather than looking like unrelated tools. Information hierarchy, modal patterns, quiet status treatments, editor proportions, and customer documents are coherent.

Area	Grade	What works	Improve next
Progress Invoice	A-	Clear scope modal, visible remaining limits, consistent editor, strong preview parity.	Clarify Draft reservation, save destination, and terminology around total versus remaining.
Change Order	B+	Estimate-style editor, clear negative rows, contextual lineage, restrained contract summaries.	Show proposed revised contract before send; surface approval evidence; reduce cognitive load in replacement groups.
Contract Invoice	A-	Source-document hierarchy, live contract progress, customer reconciliation context.	Resolve naming; make the dense six-column customer table easier to scan and print.
Cross-product	B	Shared shell and document patterns feel related.	Unify entry-point language, status semantics, Draft treatment, and contract-level navigation.

Specific UI recommendations

- Use one user-facing term for contract-wide tracked billing. The current mix of Contract Invoice, Progress Invoice, and Contract progress invoice can make contractors wonder whether these are different documents.
- Keep Contract summary operational and Contract details historical, as the prototype does. Add no more permanent dashboard blocks unless they answer a distinct question.
- In customer documents, preserve source grouping but consider print-safe repeated headers, page-break rules, and a short plain-language explanation of changed scope.
- Make reserved Draft value visible as reserved rather than folding it into Total invoiced. This is both a logic and trust issue.
- Ensure all preview, line-reference, overflow, and split-button actions have consistent hover, focus, disabled, and keyboard behavior.
- Keep destructive actions in overflow menus with explicit confirmation and a clear return destination.
- For field use, provide a focused item-selection mode with sticky totals and source headers; the full desktop table should not be the only viable interaction.

Contractor voice

I would be comfortable using these screens to prepare a draw and explain a changed contract to a customer. I would not yet trust them as the financial source of record until the system clearly distinguishes billed, reserved, paid, and remaining amounts, and tells me what happens to an open draw when a Change Order changes the contract.

9. Recommended development sequence

Phase	Deliverable	Exit condition
1. Product decisions	Approve status matrix, Draft semantics, tracked-invoice gateway, accepted-edit policy, and CO approval evidence.	No P0 item in Section 7 remains undecided.
2. Domain ledger	Contract, versioned documents, stable contract lines, invoice allocations, replacement links, and audit events.	All three prototypes can be represented without module-specific financial truth.
3. Calculation service	Accepted Contract value, posted billing, Draft reservations, remaining, over-invoiced, discount pool, and taxes.	One fixture returns identical totals in Estimate, CO, Invoice, PDF, and weblink projections.
4. Progress Invoice	Implement estimate-only tracked billing and Standard invoice copy path on the shared ledger.	First/subsequent flows, Draft lifecycle, and customer previews pass acceptance cases.
5. Change Orders	Implement creation, replacement lineage, approval lifecycle, contract updates, delete/cancel, and audit history.	Positive, negative, net-zero, multiple, and over-invoiced cases reconcile.
6. Contract Invoice	Implement contract-wide billing and transition from existing Estimate-only allocations.	Historical allocations remain stable and current eligibility reflects all Accepted COs.
7. Pilot hardening	Permissions, retainage/deposit choices, print pagination, mobile selection, accessibility, analytics, and support tooling.	A contractor pilot completes an end-to-end lifecycle without manual data repair.

Minimum review workshop agenda

- Walk one contract from Accepted Estimate through first Progress Invoice, Draft save, and Sent state.
- Accept a positive CO while no Draft exists, then confirm the transition to contract-wide invoicing.
- Repeat with an open invoice Draft and decide the collision policy.
- Replace a partially billed line and approve the historical-billing treatment.
- Accept a deductive CO that creates an over-invoiced contract and choose the v1 operational response.

- Review the same final state from Estimate, CO, internal invoice, PDF preview, and customer weblink.

Definition of review approval

This requirements set is ready for implementation planning when Product, Design, Engineering, and Finance/Operations agree on the PO decisions; each metric has one formula and one status basis; the tracked-invoice gateway is explicit; and historical issued documents remain reproducible after later contract changes.

10. Review checklist

Review question	Owner	Decision
Which invoice statuses count as billed, reserved, paid, and available?	Product + Finance	Open
Is the one-Draft rule per Estimate, per contract, or per tracked invoice type?	Product	Open
What happens to an open invoice Draft when a CO is accepted?	Product + Engineering	Open
When does the UI switch from Progress Invoice to Contract Invoice?	Product + Design	Open
May accepted Estimates or COs be edited, and how are revisions preserved?	Product + Legal/Ops	Open
How does historical billing relate to an explicit replacement line?	Finance + Product	Open
What evidence makes a CO Accepted?	Product + Legal/Ops	Open
How are retainage, deposits, credits, taxes, and discounts represented?	Finance + Product	Open
What v1 action resolves an over-invoiced contract?	Finance + Product	Open
Which customer-facing term names contract-wide tracked billing?	Design + Product	Open

Source basis

- docs/progress-invoices-prototype.md
- docs/change-orders-requirements.md
- docs/contract-invoices-prototype.md
- docs/cinderblock-ui-system.md and docs/reference-review.md
- Current React modules under src/prototypes/progress-invoices, src/prototypes/change-orders, and src/prototypes/contract-invoices
- Screenshots: creation methods, item selection, Estimate/Invoice editors, customer weblinks, and workspace references
- PDF references: estimate-1008.pdf and invoice-1007.pdf, used only as visual/document hierarchy references

Review limitations

This is a frontend product and business-logic review. It does not prescribe bookkeeping entries, tax jurisdiction rules, payment gateway behavior, QuickBooks integration, concurrency architecture, or final legal acceptance language. Those areas must be specified by the responsible domain owners before production release.

End of review